

Abdullah Alaklabi

(612) 513-4861 | alaklabi1@ksu.edu | [LinkedIn](#) | [Google Scholar](#) | [Github](#)

SUMMARY

Lecturer, Ph.D. Candidate, and Cybersecurity Researcher specializing in multimedia security and applied cryptography, with a focus on efficient image/video encryption and lightweight steganography schemes for secure data hiding.

EDUCATION

Kansas State University <i>Ph.D. in Computer Science (GPA: 3.8/4.0)</i>	<i>Manhattan, KS</i> Jan 2021 – Dec 2026
University of St. Thomas <i>M.S. in Software Engineering (GPA: 3.9/4.0)</i>	<i>Saint Paul, MN</i> Sep 2018 – Dec 2020
University of Hail <i>B.S. in Software Engineering (GPA: 3.38/4.0)</i>	<i>Saudi Arabia</i> Sep 2007 – Aug 2011

EXPERIENCE

ISCAAS Lab, Kansas State University <i>Graduate Research Assistant</i>	<i>Manhattan, KS</i> Jan 2021 – Present
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- Designed and implemented lightweight multimedia security schemes for image and video encryption, secure data hiding, and steganography using SPN-based structures, chaotic maps, and hash driven key generation.
- Developed efficient encryption and steganography pipelines leveraging SHA-256, chaotic logistic maps, LSB substitution, and confusion-diffusion mechanisms for secure multimedia protection.
- Evaluated proposed security schemes using extensive experimental analysis, including NPCR, UACI, entropy, PSNR, SSIM, histogram analysis, correlation analysis, and key sensitivity testing.

Department of Computer Science, Shaqra University <i>Lecturer</i>	<i>Saudi Arabia</i> Oct 2022 – Present
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- Appointed as Lecturer in the Department of Computer Science while continuing Ph.D. studies abroad on scholarship leave.
- Contributed to academic coordination, curriculum related tasks, and departmental responsibilities when required.

Department of Computer Science, Shaqra University <i>Teaching Assistant (On Campus until July 15, 2017; Scholarship Leave for M.S./Ph.D. Abroad)</i>	<i>Saudi Arabia</i> Jan 2014 – Oct 2022
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- Supported undergraduate instruction in core computer science courses, including Principles of Programming and Algorithms, Introduction to Databases, Data Structures, Software Engineering, and Theory of Computation.
- Delivered lab sessions and tutorials, assisted students with programming exercises, lab reports, and course projects, and provided academic support during office hours.
- Assisted faculty with grading, proctoring, preparing instructional materials and coordinating course announcements.

PROFESSIONAL SKILLS

- Multimedia Security and Applied Cryptography: Lightweight image/video encryption, steganography, secure data hiding, SPN-based encryption, chaotic logistic maps, SHA-256-based key generation, confusion-diffusion structures, LSB substitution.
- Security Evaluation and Analysis: NPCR, UACI, entropy analysis, histogram analysis, correlation analysis, key sensitivity analysis, PSNR, SSIM, FSIM, MSE, PDH, and NIST SP 800-22 randomness testing.
- Programming & Tooling: Python, MATLAB, C/C++, Linux, Git, image/video processing, experimental simulation, and data visualization.

PUBLICATIONS

- Alaklabi, A., Munir, A., Hafeez, M. A., and Khattak, M. A. K. (2024). Z-crypt: Chirp z-transform-based image encryption leveraging chaotic logistic maps and substitution permutation network. *IEEE Access*, vol. 12, pp. 123401–123422.
- Alaklabi, A., Hafeez, M. A., and Munir, A. (2026). Fast and lightweight hybrid image encryption and steganography leveraging an SPN, chaotic maps, and LSB substitution. *Journal of Cybersecurity and Privacy*, vol. 6, no. 1, p. 31.

Academic Services

President , CS Graduate Student Association (CS-GSA), K-State	2024 – 2025
Student Ambassador , K-State Libraries Student Ambassadors	2023 – 2024
Vice Chair , Department of Computer Science, Shaqra University	2016 – 2017
Media Committee Chair , Shaqra University	2015 – 2016
Second Place , Poster Presentation, University of St. Thomas	May 2019
Six Sigma Black Belt , University of St. Thomas	Jan 2018